



OPEN-ENDED LAB

CLOUD COMPUTING

SUBMITTED BY:

RABEEA FATIMA

REGISTRATION NO: 2023-BSE-049

SECTION: B

SESSION: 2023-27

SEMESTER: 5th

SUBMITTED TO:

SIR M.SHOAIB

Project Overview

This project implements a production-grade multi-tier web architecture on AWS using Infrastructure as Code (Terraform) and Configuration Management (Ansible). The system features automatic load balancing, high availability, and complete deployment automation.

Key Features:

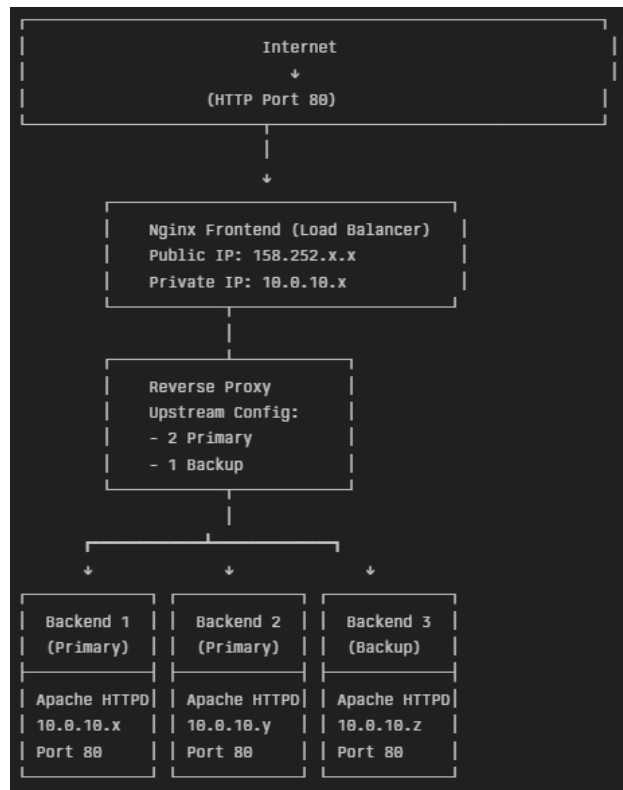
- Automated Infrastructure Provisioning: Single command deployment using Terraform
- Load Balancing: Nginx reverse proxy distributing traffic across multiple backends
- High Availability: Automatic failover to backup server when primary servers fail
- Configuration Management: Ansible roles for consistent, repeatable configuration
- Infrastructure as Code: Fully version-controlled, reproducible infrastructure
- Zero-Downtime Architecture: Backend failures don't affect service availability

Technologies Used

Component	Technology
Infrastructure	Terraform, AWS (VPC, EC2, Security Groups)
Configuration	Ansible (Roles, Playbooks, Templates)
Load Balancer	Nginx (reverse proxy with upstream)
Web Servers	Apache HTTPD
OS	Amazon Linux 2023
Automation	Terraform null_resource, Bash scripting

Project Architecture

High-Level Architecture Diagram



Project Repository

https://github.com/RabeeaFatima14/CC_RabeeaFatima_049_LabProject

CC_RabeeaFatima_049_LabProject (Public)

main 1 Branch 0 Tags

Go to file Add file Code

RabeeaFatima14 Cleanup: Add final documentation and verify project completion 8729e48 · 22 minutes ago 7 Commits

- ansible Cleanup: Add final documentation and verify project comple... 22 minutes ago
- modules/subnet Task 1: Add VPC, Subnet, IGW, Route Table, Security Group 2 hours ago
- .gitignore Initial commit: project structure and gitignore 3 hours ago
- Lab-Project-Frontend-Backend-Nginx-HA.md Cleanup: Add final documentation and verify project comple... 22 minutes ago
- README.md Task 6: Add main site.yaml playbook with dynamic backend L... 1 hour ago
- generate_inventory.sh Cleanup: Add final documentation and verify project comple... 22 minutes ago
- locals.tf Task 1: Add VPC, Subnet, IGW, Route Table, Security Group 2 hours ago
- main.tf Cleanup: Add final documentation and verify project comple... 22 minutes ago
- outputs.tf Task 2: Add frontend and backend EC2 instances 2 hours ago
- variables.tf Task 1: Add VPC, Subnet, IGW, Route Table, Security Group 2 hours ago

About

No description, website, or topics provided.

Readme Activity 0 stars 0 watching 0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
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Languages

Project:

TASK 0:

GitHub CLI Codespace Setup & Authentication.

```
C:\Users\Hp>gh codespace create --repo RabeeaFatima14/Lab15
  Codespaces usage for this repository is paid for by RabeeaFatima14
? Choose Machine Type: 2 cores, 8 GB RAM, 32 GB storage
organic-fiesta-7vj9q6694r65fq7v

C:\Users\Hp>gh codespace list
NAME                DISPLAY NAME      REPOSITORY          BRANCH  STATE      CREATED AT
jubilant-doodle-7vj9q... jubilant doodle   RabeeaFatima14/Lab9  main*   Shutdown   about 21 days ago
vigilant-meme-4jp7455... vigilant meme      RabeeaFatima14/Lab10 main*   Shutdown   about 20 days ago
supreme-waffle-97g9vr... supreme waffle    RabeeaFatima14/Assig... main*   Shutdown   about 15 days ago
zany-rotary-phone-v6j... zany rotary-phone RabeeaFatima14/Lab11 main*   Shutdown   about 13 days ago
zany-space-yodel-pj5r... zany space yodel  RabeeaFatima14/Lab12 main     Shutdown   about 11 days ago
reimagined-space-gogg... reimagined space gog... RabeeaFatima14/Lab13 main     Shutdown   about 11 days ago
solid-acorn-v6jv499v7... solid acorn       RabeeaFatima14/terra... main*   Shutdown   about 11 days ago
organic-fiesta-7vj9q6... organic fiesta    RabeeaFatima14/Lab15 main     Provisioning less than a minute ago

C:\Users\Hp>gh codespace ssh -c organic-fiesta-7vj9q6694r65fq7v
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

@RabeeaFatima14 /workspaces/Lab15 (main) $
```

TASK 01:

Installing requirements and setting up the architecture

1. Install and check the requirements

```
@RabeeaFatima14 /workspaces/Lab15 (main) $ aws --version
aws-cli/2.33.2 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
```

```
@RabeeaFatima14 /workspaces/Lab15 (main) $ aws sts get-caller-identity
{
  "UserId": "AIDAXHUW46YMRMMHDH67S",
  "Account": "497458935321",
  "Arn": "arn:aws:iam::497458935321:user/Rabeea"
}
```

```
@RabeeaFatima14 /workspaces/Lab15 (main) $ terraform --version
Terraform v1.14.3
on linux_amd64
```

```
@RabeeaFatima14 /workspaces/Lab15 (main) $ ansible --version
ansible [core 2.20.1]
  config file = None
  configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/local/py-utils/venvs/ansible-core/lib/python3.12/site-packages/ansible
  ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/local/py-utils/bin/ansible
  python version = 3.12.1 (main, Nov 27 2025, 10:47:52) [GCC 13.3.0] (/usr/local/py-utils/venvs/ansible-core/bin/python)
  Jinja version = 3.1.6
  PyYAML version = 6.0.3 (with libyaml v0.2.5)
```

2. Getting the SSH key

```
@RabeeaFatima14 /workspaces/Lab15 (main) $ ls -la ~/.ssh/
total 24
drwxr-xr-x 2 codespace codespace 4096 Jan 18 08:05 .
drwxr-xr-x 1 codespace codespace 4096 Jan 18 08:04 ..
-rw-r--r-- 1 codespace codespace 163 Jan 18 07:49 authorized_keys
-rw----- 1 codespace codespace 419 Jan 18 08:05 id_ed25519
-rw-r--r-- 1 codespace codespace 109 Jan 18 08:05 id_ed25519.pub
```

3. Creating project structure

```
@RabeeaFatima14 /workspaces/Lab15 (main) $ tree -L 3
├── Lab-Project-Frontend-Backend-Nginx-HA.md
├── README.md
├── ansible
│   ├── ansible.cfg
│   ├── inventory
│   │   └── hosts
│   ├── playbooks
│   └── site.yaml
├── roles
├── locals.tf
├── main.tf
├── modules
│   ├── subnet
│   └── webserver
├── outputs.tf
├── terraform.tfvars
└── variables.tf

8 directories, 10 files
```

4. Creating .gitignore

```
@RabeeaFatima14 /workspaces/Lab15 (main) $ @RabeeaFatima14 /workspaces/Lab15 (main) $ cat .gitignore
# Terraform
.terraform/
*.tfstate
*.tfstate.*
*.tfvars
.terraform.lock.hcl

# SSH Keys
*.pem
.ssh/id_*
!.ssh/id_*.pub

# AWS Credentials
.aws/

# Python
__pycache__/
*.pyc

# Ansible
*.retry

# OS
.DS_Store
Thumbs.db
```

5. Initial commit

```

@RabeeaFatima14 @ /workspaces/Lab15 (main) $ git add .
@RabeeaFatima14 @ /workspaces/Lab15 (main) $ git commit -m "Initial commit: project structure and gitignore"
[main 5565a40] Initial commit: project structure and gitignore
 9 files changed, 26 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 Lab-Project-Frontend-Backend-Nginx-HA.md
 create mode 100644 ansible/ansible.cfg
 create mode 100644 ansible/inventory/hosts
 create mode 100644 ansible/playbooks/site.yaml
 create mode 100644 locals.tf
 create mode 100644 main.tf
 create mode 100644 outputs.tf
 create mode 100644 variables.tf
@RabeeaFatima14 @ /workspaces/Lab15 (main) $ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (7/7), 819 bytes | 819.00 KiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/RabeeaFatima14/Lab15
   0e354fe..5565a40  main -> main

```

TASK 02:

Terraform networking infrastructure

1. Create variables.tf

```

@RabeeaFatima14 @ /workspaces/Lab15 (main) $ cat variables.tf
variable "vpc_cidr_block" {
  description = "CIDR block for VPC"
  type        = string
}

variable "subnet_cidr_block" {
  description = "CIDR block for public subnet"
  type        = string
}

variable "availability_zone" {
  description = "Availability zone for subnet"
  type        = string
}

variable "env_prefix" {
  description = "Environment prefix for resource naming"
  type        = string
}

variable "instance_type" {
  description = "EC2 instance type"
  type        = string
}

variable "public_key" {

```

2. Create terraform.tfvars

```

@RabeeaFatima14 @ /workspaces/Lab15 (main) $ cat terraform.tfvars
vpc_cidr_block      = "10.0.0.0/16"
subnet_cidr_block   = "10.0.10.0/24"
availability_zone    = "me-central-1a"
env_prefix          = "dev"
instance_type        = "t3.micro"
public_key           = "~/ssh/id_ed25519.pub"
private_key          = "~/ssh/id_ed25519"

```

3. Create locals.tf

```
@RabeeaFatima14 [ ] /workspaces/Lab15 (main) $ cat locals.tf
data "http" "my_ip" {
  url = "https://api.ipify.org?format=text"
}

locals {
  my_ip = "${chomp(data.http.my_ip.response_body)}/32"
}
```

4. Create Subnet Module

```
@RabeeaFatima14 [ ] /workspaces/Lab15 (main) $ tree modules/subnet/
modules/subnet/
├── main.tf
├── outputs.tf
└── variables.tf

1 directory, 3 files
```

5. Create Main Terraform Configuration

```
@RabeeaFatima14 [ ] /workspaces/Lab15 (main) $ nano main.tf
@RabeeaFatima14 [ ] /workspaces/Lab15 (main) $ head -n 30 main.tf
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.0"
    }
    http = {
      source = "hashicorp/http"
      version = "~> 3.0"
    }
  }
}

provider "aws" {
  region = "me-central-1"
}

# VPC
resource "aws_vpc" "myapp_vpc" {
  cidr_block      = var.vpc_cidr_block
  enable_dns_support = true
  enable_dns_hostnames = true

  tags = {
```

6. Initialize Terraform

```
@RabeeaFatima14 [ ] /workspaces/Lab15 (main) $ terraform init
Initializing the backend...
Initializing modules...
- myapp-subnet in modules/subnet
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Finding hashicorp/http versions matching "~> 3.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

7. Validate Configuration

```
@RabeeaFatima14 [ ] /workspaces/Lab15 (main) $ terraform validate
Success! The configuration is valid.
```

8. Test Plan

```
@RabeeaFatima14 [ ] /workspaces/Lab15 (main) $ terraform plan
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 1s [id=https://api.ipify.org?format=text]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

# aws_security_group.myapp_sg will be created
+ resource "aws_security_group" "myapp_sg" {
+   arn                = (known after apply)
+   description        = "Security group for frontend and backend servers"
+   egress              = [
+     {
+       cidr_blocks     = [
+         "0.0.0.0/0",
+       ]
+       from_port        = 0
+       ipv6_cidr_blocks = []
+       prefix_list_ids  = []
+       protocol         = "-1"
+       security_groups  = []
+       self              = false
+     }
+   ]
+ }
```

9. Commit Networking Code

```

@RabeeaFatima14 [ /workspaces/Lab15 (main) ] $ git add variables.tf terraform.tfvars locals.tf main.tf modules/subnet/
The following paths are ignored by one of your .gitignore files:
terraform.tfvars
hint: Use -f if you really want to add them.
hint: Disable this message with "git config set advice.addIgnoredFile false"
@RabeeaFatima14 [ /workspaces/Lab15 (main) ] $ git commit -m "Task 1: Add VPC, Subnet, IGW, Route Table, Security Group"
[main ec1381c] Task 1: Add VPC, Subnet, IGW, Route Table, Security Group
6 files changed, 179 insertions(+)
create mode 100644 modules/subnet/main.tf
create mode 100644 modules/subnet/outputs.tf
create mode 100644 modules/subnet/variables.tf
@RabeeaFatima14 [ /workspaces/Lab15 (main) ] $ git push origin main
Enumerating objects: 14, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 2 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (10/10), 1.87 KiB | 957.00 KiB/s, done.
Total 10 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/RabeeaFatima14/Lab15
5565a40..ec1381c main -> main

```

TASK 03:

Terraform EC2 Instances

1. Get Latest Amazon Linux 2023 AMI

```

@RabeeaFatima14 [ /workspaces/Lab15 (main) ] $ aws ec2 describe-images \
> --region me-central-1 \
> --owners amazon \
> --filters "Name=name,Values=al2023-ami-2023*" "Name=architecture,Values=x86_64" \
> --query 'Images | sort_by(@, &CreationDate) | [-1].ImageId' \
> --output text
ami-07aca7a231992279f

```

2. Create SSH Key Pair Resource

```

# SSH Key Pair
resource "aws_key_pair" "myapp_key" {
  key_name   = "${var.env_prefix}-key"
  public_key = file(var.public_key)

  tags = {
    Name = "${var.env_prefix}-key"
  }
}

```

3. Add Frontend EC2 Instance

```

# Frontend EC2 Instance (Nginx)
resource "aws_instance" "frontend" {
  ami                  = "ami-07aca7a231992279f"
  instance_type        = var.instance_type
  subnet_id            = module.myapp-subnet.subnet.id
  vpc_security_group_ids = [aws_security_group.myapp_sg.id]
  availability_zone     = var.availability_zone
  key_name              = aws_key_pair.myapp_key.key_name

  associate_public_ip_address = true

  tags = {
    Name = "${var.env_prefix}-frontend"
    Type = "frontend"
    Role = "nginx"
  }
}

```

4. Add Backend EC2 Instances

```
# Backend EC2 Instances (HTTPD) - 3 instances
resource "aws_instance" "backend" {
  count                = 3
  ami                 = "ami-07aca7a231992279f"
  instance_type       = var.instance_type
  subnet_id           = module.myapp-subnet.subnet.id
  vpc_security_group_ids = [aws_security_group.myapp_sg.id]
  availability_zone    = var.availability_zone
  key_name             = aws_key_pair.myapp_key.key_name

  associate_public_ip_address = true

  tags = {
    Name = "${var.env_prefix}-backend-${count.index + 1}"
    Type = "backend"
    Role = "httpd"
  }
}
```

5. Create Outputs File

```
@RabeeaFatima14 [ ] /workspaces/Lab15 (main) $ cat outputs.tf
# Frontend Outputs
output "frontend_public_ip" {
  description = "Public IP of frontend nginx server"
  value       = aws_instance.frontend.public_ip
}

output "frontend_private_ip" {
  description = "Private IP of frontend nginx server"
  value       = aws_instance.frontend.private_ip
}

# Backend Outputs
output "backend_public_ips" {
  description = "Public IPs of backend httpd servers"
  value       = [for instance in aws_instance.backend : instance.public_ip]
}

output "backend_private_ips" {
  description = "Private IPs of backend httpd servers"
  value       = [for instance in aws_instance.backend : instance.private_ip]
}

# All IPs for inventory generation
output "all_backend_info" {
  description = "All backend instance information"
  value       = aws_instance.backend
}
```

6. Validate Configuration

```
@RabeeaFatima14 [ ] /workspaces/Lab15 (main) $ terraform validate
Success! The configuration is valid.
```

7. Plan infrastructure

```

+ private_dns_hostname_type_on_launch = (known after apply)
+ tags                                 = {
  + "Name" = "dev-subnet-1"
}
+ tags_all                             = {
  + "Name" = "dev-subnet-1"
}
+ vpc_id                               = (known after apply)
}

```

Plan: 10 to add, 0 to change, 0 to destroy.

8. Apply infrastructure

```

aws_instance.backend[2]: Still creating... [00m10s elapsed]
aws_instance.backend[1]: Still creating... [00m10s elapsed]
aws_instance.backend[0]: Still creating... [00m10s elapsed]
aws_instance.frontend: Still creating... [00m10s elapsed]
aws_instance.backend[1]: Creation complete after 13s [id=i-0e2ac952451d43dc7]
aws_instance.backend[2]: Creation complete after 13s [id=i-0651faec217b405ee]
aws_instance.backend[0]: Creation complete after 13s [id=i-0ca0604f963302221]
aws_instance.frontend: Creation complete after 13s [id=i-05431373277ef3592]

```

Apply complete! Resources: 10 added, 0 changed, 0 destroyed.

9. Output

```

all_backend_info = [
  {
    "name" = "backend-1"
    "private_ip" = "10.0.10.94"
    "public_ip" = "40.172.234.126"
  },
  {
    "name" = "backend-2"
    "private_ip" = "10.0.10.149"
    "public_ip" = "51.112.45.77"
  },
  {
    "name" = "backend-3"
    "private_ip" = "10.0.10.121"
    "public_ip" = "51.112.167.221"
  },
]
backend_private_ips = [
  "10.0.10.94",
  "10.0.10.149",
  "10.0.10.121",
]
backend_public_ips = [
  "40.172.234.126",
  "51.112.45.77",
  "51.112.167.221",
]
frontend_private_ip = "10.0.10.240"
frontend_public_ip = "3.29.232.6"

```

10. Test SSH Access to Frontend

[illegible]

TASK 04: Ansible Setup

1. Create Ansible Directory Structure

```

@RabeeaFatima14 /workspaces/Lab15/ansible (main) $ tree -L 2 roles/
roles/
├── backend
│   ├── README.md
│   ├── defaults
│   ├── files
│   ├── handlers
│   ├── meta
│   ├── tasks
│   ├── templates
│   ├── tests
│   └── vars
└── frontend
    ├── README.md
    ├── defaults
    ├── files
    ├── handlers
    ├── meta
    ├── tasks
    ├── templates
    ├── tests
    └── vars

```

19 directories, 2 files

2. Create ansible.cfg

```
@RabeeaFatima14 ▢ /workspaces/Lab15/ansible (main) $ cat ansible.cfg
[defaults]
host_key_checking = False
interpreter_python = /usr/bin/python3
inventory = ./inventory/hosts
roles_path = ./roles
deprecation_warnings = False

[ssh_connection]
pipelining = True
```

3. Create ansible.cfg

```
@RabeeaFatima14 [ ] /workspaces/Lab15/ansible (main) $ cat ansible.cfg
[defaults]
host_key_checking = False
interpreter_python = /usr/bin/python3
inventory = ./inventory/hosts
roles_path = ./roles
deprecation_warnings = False

[ssh_connection]
pipelining = True
```

4. Create ansible inventory file

```
[frontend]
3.29.232.6

[backends]
40.172.234.126
51.112.45.77
51.112.167.221

[all:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_python_interpreter=/usr/bin/python3
```

5. Test Ansible Connectivity to Frontend

```
@RabeeaFatima14 [ ] /workspaces/Lab15/ansible (main) $ ansible frontend -m ping
3.29.232.6 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

6. Test Ansible Connectivity to All Backends

```
@RabeeaFatima14 [ ] /workspaces/Lab15/ansible (main) $ ansible backends -m ping
40.172.234.126 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
51.112.167.221 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
51.112.45.77 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

7. Verify Ansible Can See Groups

```
@RabeeaFatima14 [ ] /workspaces/Lab15/ansible (main) $ ansible-inventory --graph
@all:
|--@ungrouped:
|--@frontend:
| |--3.29.232.6
|--@backends:
| |--40.172.234.126
| |--51.112.45.77
| |--51.112.167.221
```

TASK 05:

Backend Ansible Role (HTTPD Installation + Distinct Content)

1. Create Test Playbook for Backend Only

```
@RabeeaFatima14 [ /workspaces/Lab15/ansible (main) ] $ ansible-playbook playbooks/test_backend.yaml

PLAY [Test Backend Role] *****

TASK [Gathering Facts] *****
ok: [51.112.167.221]
ok: [51.112.45.77]
ok: [40.172.234.126]

TASK [backend : Install httpd] *****
changed: [51.112.167.221]
changed: [51.112.45.77]
changed: [40.172.234.126]

TASK [backend : Enable and start httpd service] *****
changed: [51.112.45.77]
changed: [40.172.234.126]
changed: [51.112.167.221]

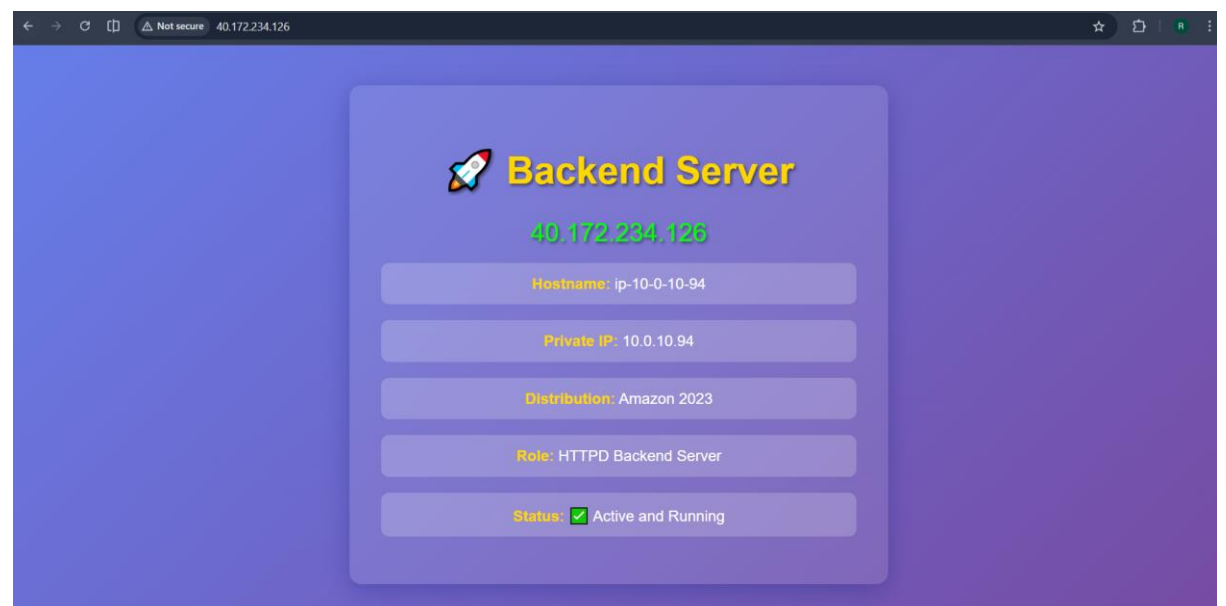
TASK [backend : Deploy backend index page] *****
changed: [51.112.45.77]
changed: [40.172.234.126]
changed: [51.112.167.221]

RUNNING HANDLER [backend : Restart httpd] *****
changed: [51.112.45.77]
changed: [40.172.234.126]
changed: [51.112.167.221]

PLAY RECAP *****
40.172.234.126      : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
51.112.167.221    : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
51.112.45.77      : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

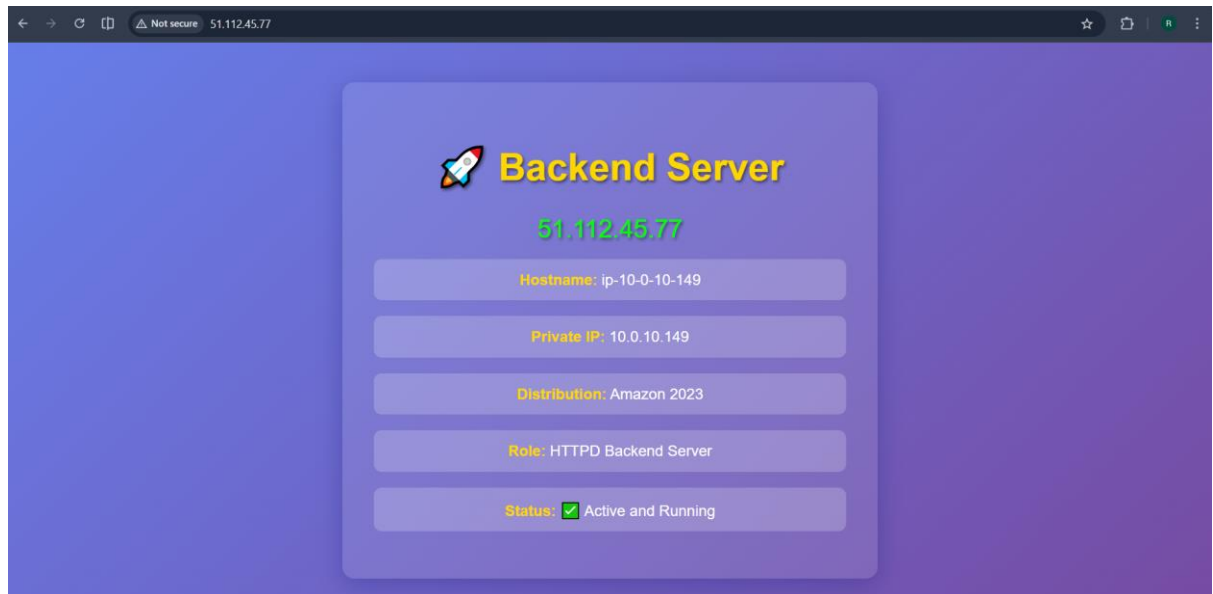
2. Test Backend 1 Directly in Browser

```
@RabeeaFatima14 [ /workspaces/Lab15/ansible (main) ] $ cd /workspaces/Lab15
@RabeeaFatima14 [ /workspaces/Lab15 (main) ] $ terraform output -json backend_public_ips | jq -r '.[0]'
40.172.234.126
```



3. Test Backend 2 Directly in Browser

```
@RabeeaFatima14 /workspaces/Lab15 (main) $ terraform output -json backend_public_ips | jq -r '[1]'
51.112.45.77
```



4. Backend role structure

```
@RabeeaFatima14 /workspaces/Lab15/ansible (main) $ tree roles/backend/
roles/backend/
├── README.md
├── defaults
│   └── main.yml
├── files
├── handlers
│   └── main.yml
├── meta
│   └── main.yml
├── tasks
│   └── main.yml
├── templates
│   └── backend_index.html.j2
├── tests
│   ├── inventory
│   └── test.yml
└── vars
    └── main.yml

9 directories, 9 files
```

TASK 06:

Ansible Roles – Frontend Nginx Role (frontend/)

1. Run Frontend Role

```
@RabeeaFatima14 /workspaces/Lab15/ansible (main) $ ansible-playbook playbooks/test_frontend.yaml

PLAY [Test Frontend Role] *****

TASK [Gathering Facts] *****
ok: [3.29.232.6]

TASK [frontend : Install nginx] *****
changed: [3.29.232.6]

TASK [frontend : Deploy nginx configuration] *****
changed: [3.29.232.6]

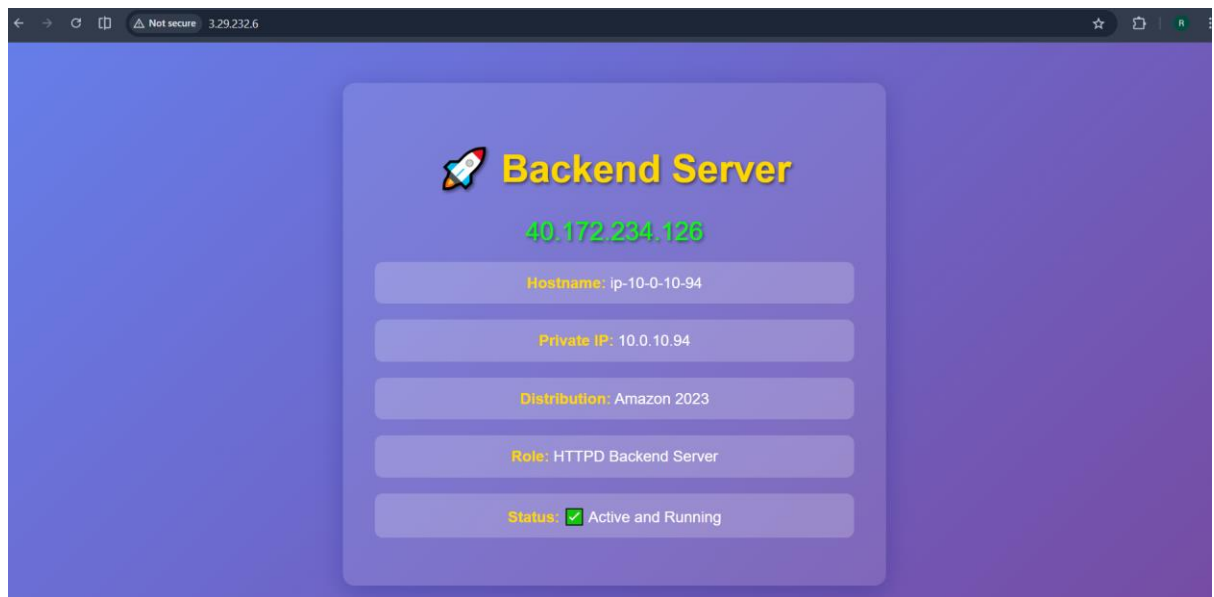
TASK [frontend : Enable and start nginx] *****
changed: [3.29.232.6]

RUNNING HANDLER [frontend : Restart nginx] *****
changed: [3.29.232.6]

PLAY RECAP *****
3.29.232.6 : ok=5 changed=4 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
```

2. Test in Browser

```
@RabeeaFatima14 /workspaces/Lab15/ansible (main) $ cd /workspaces/Lab15
@RabeeaFatima14 /workspaces/Lab15 (main) $ terraform output frontend_public_ip
"3.29.232.6"
```



3. Verify Nginx Access Logs

```
@RabeeaFatima14 /workspaces/Lab15 (main) $ ssh -i ~/.ssh/id_ed25519 ec2-user@$(terraform output -raw frontend_public_ip)

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Sun Jan 18 09:21:33 2026 from 20.192.21.53
[ec2-user@ip-10-0-10-240 ~]$ sudo tail -n 20 /var/log/nginx/access.log
182.177.77.170 - - [18/Jun/2026:10:05:53 +0000] "GET / HTTP/1.1" 200 1902 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" upstream: 10.0.10.94:80
182.177.77.170 - - [18/Jun/2026:10:05:53 +0000] "GET /favicon.ico HTTP/1.1" 404 236 "http://3.29.232.6/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" upstream: 10.0.10.149:80
[ec2-user@ip-10-0-10-240 ~]$ exit
logout
Connection to 3.29.232.6 closed.
```

4. Test Upstream Configuration

```
[ec2-user@ip-10-0-10-240 ~]$ sudo cat /etc/nginx/nginx.conf | grep -A 5 "upstream backend_servers"
    upstream backend_servers {
        server 10.0.10.94:80;
        server 10.0.10.149:80;
        server 10.0.10.121:80 backup;
    }
```

5. Verify Frontend Role Structure

```
@RabeeaFatima14 [ /workspaces/Lab15/ansible (main) ] $ tree roles/frontend/
roles/frontend/
├── README.md
├── defaults
│   └── main.yml
├── files
├── handlers
│   └── main.yml
├── meta
│   └── main.yml
├── tasks
│   └── main.yml
├── templates
│   └── nginx_frontend.conf.j2
├── tests
│   ├── inventory
│   └── test.yml
└── vars
    └── main.yml

9 directories, 9 files
```

TASK 07:

Ansible Main Playbook Using Roles

1. Run Complete Playbook

```
@RabeeaFatima14 [ /workspaces/Lab15/ansible (main) ] $ ansible-playbook playbooks/site.yml

PLAY [Configure Backend HTTPD Servers] *****

TASK [Gathering Facts] *****
ok: [51.112.45.77]
ok: [40.172.234.126]
ok: [51.112.167.221]

TASK [backend : Install httpd] *****
ok: [51.112.167.221]
ok: [40.172.234.126]
ok: [51.112.45.77]

TASK [backend : Enable and start httpd service] *****
ok: [51.112.45.77]
ok: [51.112.167.221]
ok: [40.172.234.126]

TASK [backend : Deploy backend index page] *****
ok: [51.112.45.77]
ok: [40.172.234.126]
ok: [51.112.167.221]

PLAY [Configure Frontend Nginx Load Balancer] *****

TASK [Gathering Facts] *****
ok: [3.29.232.6]

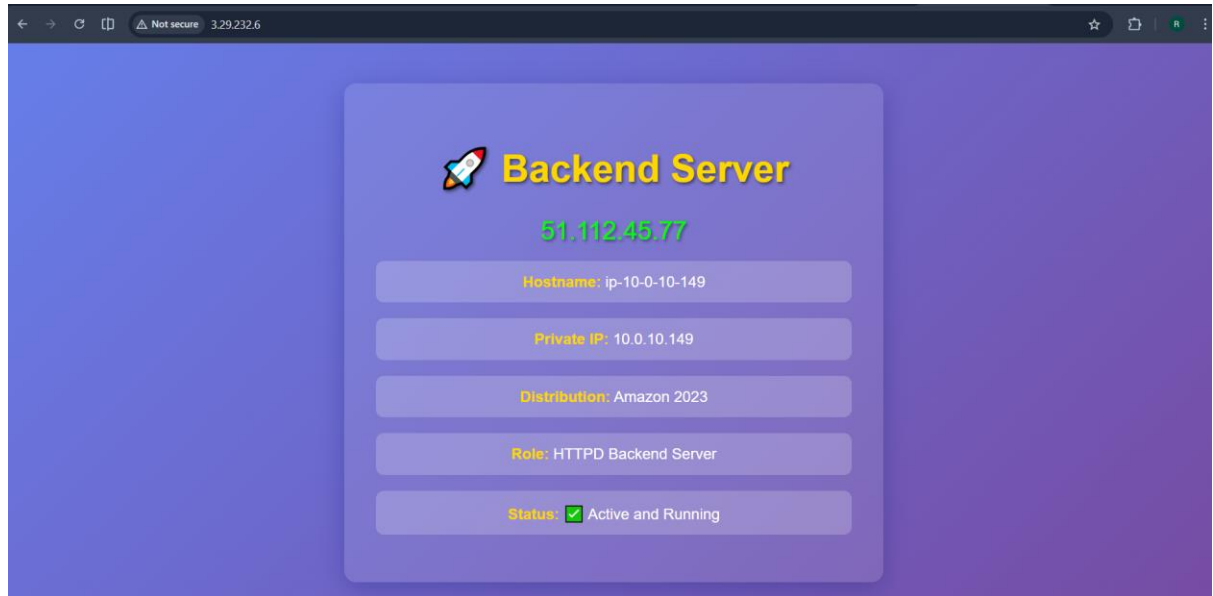
TASK [frontend : Install nginx] *****
ok: [3.29.232.6]

TASK [frontend : Deploy nginx configuration] *****
ok: [3.29.232.6]

TASK [frontend : Enable and start nginx] *****
ok: [3.29.232.6]

PLAY RECAP *****
3.29.232.6      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
40.172.234.126 : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
51.112.167.221 : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
51.112.45.77   : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

2. Test in Browser



3. Verify Nginx Logs Show Upstream Info

```
@RabeeaFatima14: /workspaces/Lab15 (main) $ ssh -i ~/.ssh/id_ed25519 ec2-user@(terraform output -raw frontend_public_ip)
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Sun Jan 18 10:15:58 2026 from 20.192.21.53
ec2-user@ip-10-0-10-240 ~$ sudo tail -n 20 /var/log/nginx/access.log
185.16.39.146 - - [18/Jan/2026:10:07:43 +0000] "GET / HTTP/1.1" 200 1902 "-" "wget" "-" upstream: 10.0.10.94:80
20.192.21.53 - - [18/Jan/2026:10:09:45 +0000] "GET / HTTP/1.1" 200 1904 "-" "curl/8.5.0" "-" upstream: 10.0.10.149:80, 10.0.10.94:80, 10.0.10.121:80
20.192.21.53 - - [18/Jan/2026:10:10:08 +0000] "GET / HTTP/1.1" 200 1902 "-" "curl/8.5.0" "-" upstream: 10.0.10.94:80
20.192.21.53 - - [18/Jan/2026:10:10:10 +0000] "GET / HTTP/1.1" 200 1902 "-" "curl/8.5.0" "-" upstream: 10.0.10.94:80
20.192.21.53 - - [18/Jan/2026:10:10:12 +0000] "GET / HTTP/1.1" 200 1900 "-" "curl/8.5.0" "-" upstream: 10.0.10.149:80
20.192.21.53 - - [18/Jan/2026:10:10:14 +0000] "GET / HTTP/1.1" 200 1902 "-" "curl/8.5.0" "-" upstream: 10.0.10.94:80
20.192.21.53 - - [18/Jan/2026:10:16:18 +0000] "GET / HTTP/1.1" 200 1900 "-" "curl/8.5.0" "-" upstream: 10.0.10.149:80
20.192.21.53 - - [18/Jan/2026:10:16:20 +0000] "GET / HTTP/1.1" 200 1902 "-" "curl/8.5.0" "-" upstream: 10.0.10.94:80
20.192.21.53 - - [18/Jan/2026:10:16:22 +0000] "GET / HTTP/1.1" 200 1900 "-" "curl/8.5.0" "-" upstream: 10.0.10.149:80
20.192.21.53 - - [18/Jan/2026:10:16:23 +0000] "GET / HTTP/1.1" 200 1902 "-" "curl/8.5.0" "-" upstream: 10.0.10.94:80
20.192.21.53 - - [18/Jan/2026:10:16:25 +0000] "GET / HTTP/1.1" 200 1900 "-" "curl/8.5.0" "-" upstream: 10.0.10.149:80
20.192.21.53 - - [18/Jan/2026:10:16:27 +0000] "GET / HTTP/1.1" 200 1902 "-" "curl/8.5.0" "-" upstream: 10.0.10.94:80
20.192.21.53 - - [18/Jan/2026:10:16:29 +0000] "GET / HTTP/1.1" 200 1900 "-" "curl/8.5.0" "-" upstream: 10.0.10.149:80
20.192.21.53 - - [18/Jan/2026:10:16:31 +0000] "GET / HTTP/1.1" 200 1902 "-" "curl/8.5.0" "-" upstream: 10.0.10.94:80
192.177.77.170 - - [18/Jan/2026:10:17:10 +0000] "GET / HTTP/1.1" 200 1900 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36" "-" upstream: 10.0.10.149:80
185.16.39.146 - - [18/Jan/2026:10:22:44 +0000] "GET / HTTP/1.1" 200 1902 "-" "wget" "-" upstream: 10.0.10.94:80
20.192.21.53 - - [18/Jan/2026:10:24:06 +0000] "GET / HTTP/1.1" 200 1900 "-" "curl/8.5.0" "-" upstream: 10.0.10.149:80
20.192.21.53 - - [18/Jan/2026:10:24:08 +0000] "GET / HTTP/1.1" 200 1900 "-" "curl/8.5.0" "-" upstream: 10.0.10.94:80, 10.0.10.149:80
20.192.21.53 - - [18/Jan/2026:10:24:09 +0000] "GET / HTTP/1.1" 200 1900 "-" "curl/8.5.0" "-" upstream: 10.0.10.149:80
20.192.21.53 - - [18/Jan/2026:10:24:11 +0000] "GET / HTTP/1.1" 200 1900 "-" "curl/8.5.0" "-" upstream: 10.0.10.149:80
[ec2-user@ip-10-0-10-240 ~]$ exit
logout
Connection to 3.29.232.6 closed.
```

4. Verify Complete Playbook Structure

```
@RabeeaFatima14: /workspaces/Lab15/ansible (main) $ tree playbooks/
playbooks/
├── show_backend_vars.yaml
├── site.yaml
├── test_backend.yaml
└── test_frontend.yaml

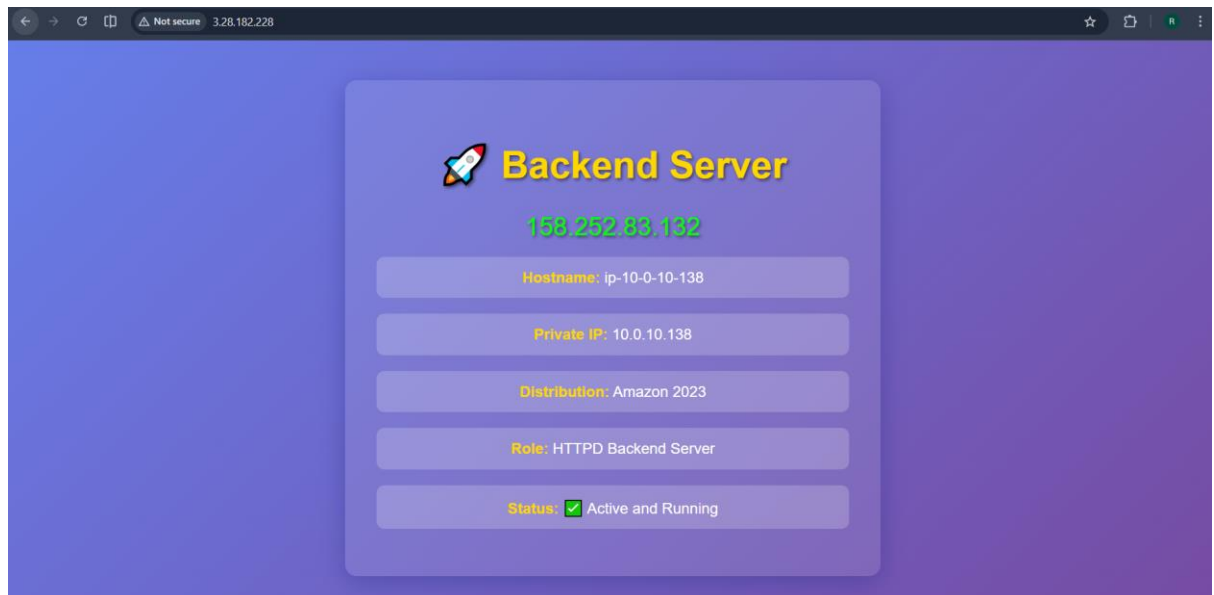
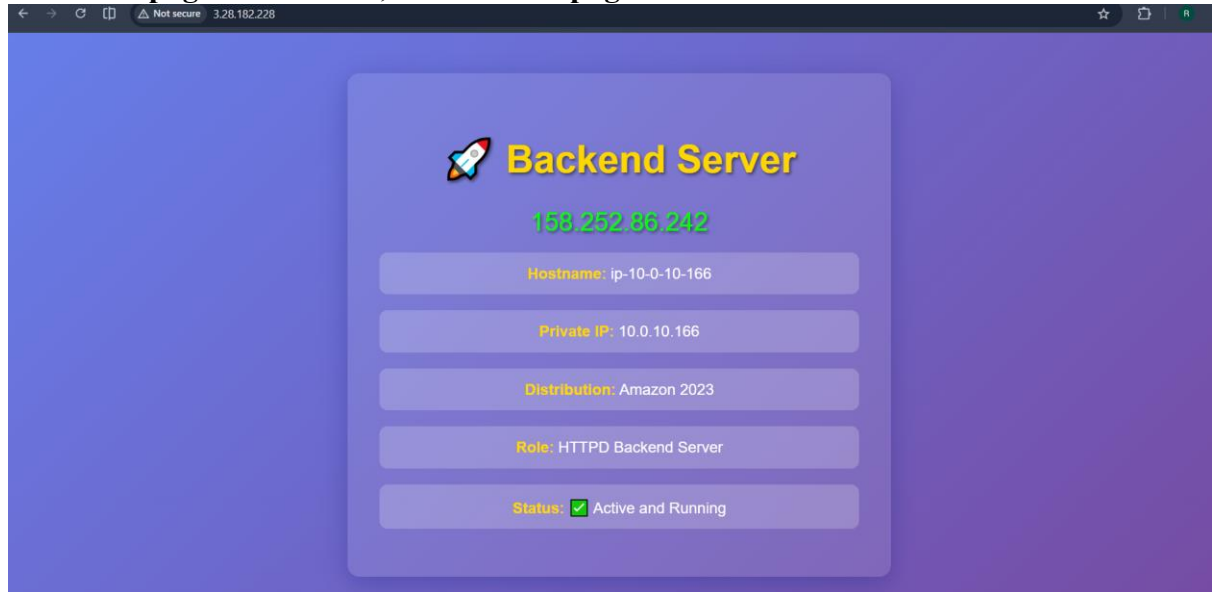
1 directory, 4 files
```

TASK 08:

Terraform–Ansible Integration

1. Testing in the browser

When the page is refreshed, the backend pages alters



2. Terraform automatically triggers and forms EC2 instances

```
null_resource.configure_servers: Creation complete after 1m18s [id=4909864152346331934]
```

```
Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
```

```
Outputs:
```

```
all_backend_info = [  
  {  
    "name" = "backend-1"  
    "private_ip" = "10.0.10.138"  
    "public_ip" = "158.252.83.132"  
  },  
  {  
    "name" = "backend-2"  
    "private_ip" = "10.0.10.166"  
    "public_ip" = "158.252.86.242"  
  },  
  {  
    "name" = "backend-3"  
    "private_ip" = "10.0.10.117"  
    "public_ip" = "51.112.49.75"  
  },  
]  
backend_private_ips = [  
  "10.0.10.138",  
  "10.0.10.166",  
  "10.0.10.117",  
]  
backend_public_ips = [  
  "158.252.83.132",  
  "158.252.86.242",  
  "51.112.49.75",  
]  
frontend_private_ip = "10.0.10.143"  
frontend_public_ip = "3.28.182.228"
```